

Introduction to Natural Language Processing (NLP)

What is Natural Language Processing?

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human language.

The goal of NLP is to allow communication between humans and computers using natural language.

Why is NLP Important?

Humans communicate using languages such as English, Urdu, and many others. Computers understand numerical data, so NLP helps convert human language into a format that machines can process.

Common NLP Tasks

1. Text Classification

Text classification assigns categories to text data.

Examples:

- Spam email detection
- Sentiment analysis
- News classification

2. Sentiment Analysis

Sentiment analysis identifies emotions or opinions expressed in text.

Example:

A product review can be classified as:

- Positive
- Negative

- Neutral

3. Machine Translation

Machine translation converts text from one language to another.

Examples:

- Google Translate
- Language translation systems

4. Chatbots

NLP allows chatbots to understand user questions and provide meaningful responses.

Examples:

- Customer support bots
- AI assistants

NLP Techniques

Tokenization

Tokenization is the process of breaking text into smaller units called tokens.

Example:

Sentence:

"Machine Learning is powerful"

Tokens:

Machine, Learning, is, powerful

Word Embeddings

Word embeddings convert words into numerical vectors that capture their meaning.

Examples:

- Word2Vec
- GloVe

Transformers

Transformers are advanced neural network architectures used in modern NLP systems.

Examples:

- BERT
- GPT models

Large Language Models (LLMs)

Large Language Models are AI models trained on massive amounts of text data.

They can:

- Answer questions
- Generate text
- Summarize documents
- Translate languages
- Write code

Examples:

- Gemini
- GPT
- Claude

Applications of NLP

NLP is used in:

Virtual Assistants

Examples:

- Voice assistants
- AI chatbots

Search Engines

NLP helps search engines understand user queries.

Healthcare

NLP analyzes medical documents and patient records.

Business

Companies use NLP for customer feedback analysis.

Conclusion

Natural Language Processing helps computers understand and communicate using human language. It is a key technology behind modern AI applications such as chatbots, search engines, and large language models.